FORWARD DEPLOYED AI ENGINEERING

Capstone Project Report

CloudServe Support Automation

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Primary evaluation: clean-checkout run on 8 November 2026, one invocation, 80 visible validation tickets

Submission target: 13 November 2026

This report describes a local prototype and a reproducible visible-set run. It is not a deployment approval. The hidden assessment dataset was not available in this workspace. The student confirmed 45.5 actual hours through 8 November; the remaining 17.5 hours are scheduled, not yet actual. Signature remains pending.

Contents and evaluation record

Main report: 20 pages, excluding appendices, followed by supporting evidence. The section sequence follows the submission guide.

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Primary run: 2026-11-08 16:09:41 UTC; system revision d285977; input SHA-256 94c5b1adc1203f06d59522a909d755e8671e9bf1b165a0bc5a74ab62501c827b. The current run processed the supplied visible validation set. No hidden ticket data were accessed.

1. Executive Summary

CloudServe requested a chatbot to help with a high-volume support queue. Discovery evidence and the supplied datasets support a narrower first build: classify one incoming ticket, retrieve approved documentation, and return a sourced draft only when policy checks pass; otherwise prepare a human handoff. The prototype exposes a FastAPI endpoint and browser chat, uses scikit-learn classifiers, TF-IDF retrieval in Chroma, deterministic validators, SQLite decision logging, and optional Groq/OpenRouter generation.

On 8 November 2026, a fresh clone of the packaged repository processed all 80 visible validation tickets in one unattended harness invocation. It returned 33 automatic responses (41.25%), escalated 47 (58.75%), and blocked zero. Intent accuracy was 100% on the supplied labels; urgency accuracy was 43.75%. Expected-document retrieval hit 49 of 53 eligible tickets (92.45%). The route accuracy against expected-route labels was 63.75%, and intent expected calibration error was 16.64 percentage points, above the five-point target.

Measure	8 Nov clean-checkout result
Processed / decisions logged	80 / 80
Auto response / escalation / block	33 (41.25%) / 47 (58.75%) / 0
Intent / urgency accuracy	100% / 43.75%
Expected source hit rate	49/53 (92.45%)
Intent ECE / p95 local latency	16.64 pp / 3.55 ms

The sample supports technical path execution, not live FCR, customer satisfaction, service availability, or answer quality. The current package passes all 16 tests in a fresh checkout. No independent two-assessor review of 50 answers, live provider request, hidden assessment run, or production pilot was performed. Keep this as a human-supervised prototype; the figures do not establish customer benefit.

2. The Problem: Client Request and Queue Context

The fictional CloudServe brief describes a business-to-business cloud infrastructure and developer operations company handling more than 500 weekly support tickets across email, chat, documentation comments, and a community forum. It reports six agents, an 8–12 hour typical response interval, a two-hour contractual response commitment, and low first-contact resolution. The brief asks for a chatbot.

The request describes an interface, while the operational problem spans discovery, response quality, routing, and accountability. A chat window could produce answers quickly while still increasing repeat contacts if it retrieves the wrong policy, misses urgency, or states unsupported steps. The product framing therefore includes decision support and a controlled handoff, not only natural-language generation.

Stakeholder transcripts are provided as fictional capstone evidence. They describe poor search, differences between familiar and unusual ticket handling time, weak knowledge titles, unsafe reuse of private notes, and a customer preference for timely answers with clear sources and uncertainty. These perspectives are useful as supplied evidence but are not interviews personally conducted by the student.

A safe initial user journey is: submit one ticket; normalize channel and metadata; predict intent and urgency; retrieve approved passages; apply policy and confidence gates; optionally draft a response; validate it; then return either an answer or an explicit human-review outcome. A batch harness reuses this same path for evaluation. The API remains one-ticket-per-request; batch input is an evaluation facility, not multi-ticket conversational state.

2. The Problem: Baselines, Targets, and Scope

The development data contain 500 labeled synthetic tickets. Of these, 219 (43.8%) have a historical first-contact-resolution label, 108 (21.6%) have a repeat-contact label, and mean historical customer satisfaction is 2.972/5. These data fields are dataset baselines. They are not telemetry from a CloudServe production deployment and cannot show that this prototype changes customer outcomes.

The project instructions and evaluation framework identify FCR of at least 60%, CSAT of at least 4.0/5, and a first automated reply under five minutes as business goals; a separate Project Brief variant mentions 65% FCR and 4.2 CSAT. The report treats the main project instructions and evaluation framework as the assessment target, while keeping the inconsistency visible for course-owner clarification. The contractual two-hour response promise measures a separate service clock.

In scope	Out of scope
Ticket normalization, prediction, retrieval, sourced draft, escalation, guardrails, logs, offline evaluation.	Autonomous account changes, unsupervised security handling, replacing agents, or claims of a production launch.
Measurable prototype behavior and a human-supervised next-stage plan.	Claims that offline latency equals customer response time or automatic response equals FCR.

The project should be assessed on safe, explainable handling and reproducible evidence. Queue reduction is only a hypothesis until the organization measures response timestamps, resolution, reopens, customer ratings, and agent workload under a controlled pilot.

3. Discovery Findings: Stakeholder Evidence

The supplied pack contains five stakeholder transcripts. The Head of Support describes queue volume above 500 weekly tickets, six agents, an 8–12 hour response time against a two-hour commitment, and escalation cost estimated at roughly four times a first-line resolution. The Tier One agent describes 40–70 tickets daily, four to five minutes for familiar questions, and up to 40 minutes for unusual cases; search quality and use of private snippets are concerns.

The Tier Two engineer estimates that some escalations could be resolved by Tier One when relevant documentation and a useful context summary are available. The technical writer describes a 29-article knowledge base, weak exact-title search, and the risks of unreviewed private notes. The customer asks for timely answers that make sources and uncertainty visible.

These accounts point to three design needs: retrieve approved knowledge with stable references; provide a concise reason and evidence summary when handing a ticket to a human; and avoid automating decisions that require account authority or specialist judgment. The interviews do not establish causal relationships or population-level rates. They are a small qualitative input supplied for the exercise.

The system addresses the search and handoff workflow as a prototype. It does not integrate with a ticketing platform, assign a live agent, or alter customer accounts. An operational pilot would need named owners, channel integrations, privacy controls, and measured queue outcomes.

3. Discovery Findings: Dataset and Corpus

The development dataset has 500 synthetic tickets with intent, urgency, expected route, historical outcome, customer tier, region, and fluency fields. In this sample, 357 tickets (71.4%) are labeled answerable from documentation and 311 (62.2%) are marked as expected to route automatically. These labels describe the constructed sample and are not directly observable outcomes from deployed users.

Development evidence	Observed value	What it does and does not show
Ticket count	500	Supplied labeled sample
Document answerability	357 / 500 (71.4%)	Potential corpus coverage; not answer safety
Expected auto route	311 / 500 (62.2%)	Dataset route label; not system success
Historical FCR	219 / 500 (43.8%)	Baseline field; not prototype FCR
Historical repeat contact	108 / 500 (21.6%)	Baseline label; not caused by the prototype
Historical CSAT	2.972 / 5	Dataset rating; not rating of generated answers
Knowledge base	29 articles	Approved retrieval corpus for the exercise

The development sample has a different purpose from the 80-ticket visible validation set. The reference-response file contains 200 senior-agent examples matched to development ticket IDs. Reference text is an evaluation rubric, not a runtime response template; it must not leak into the validation set or be used as a production knowledge source.

3. Discovery Findings: Gaps and Design Consequences

The sample does not provide dependable customer-visible first-reply timestamps for the validation run, system resolution events, live CSAT, or agent time per task. It also does not provide independent ratings of generated responses. As a result, the evaluation can describe action shares, label-based model scores, retrieval proxies, and local computation latency, but it cannot measure business improvement or semantic answer quality.

The supplied development data show historical FCR of 45.8% for non-fluent customers and 43.2% for fluent customers, with historical CSAT of 3.04 and 2.95 respectively. The direction differs from some interview concerns. These are synthetic historic outcomes and neither establish nor rule out a disparity in automated response quality.

The latest validation run has small subgroup counts, including enterprise tier (n=8), Latin America (n=7), and non-fluent language group (n=19). Reporting subgroup coverage is useful for identifying where a larger audit is needed, but the results are descriptive and should not be treated as causal or as proof of fairness.

The resulting design keeps a human path available at each uncertain step. Evidence-sensitive claims are blocked by deterministic checks; elevated-risk intents are routed to people; local decision records support review. Before pilot use, the team would need lifecycle timestamps, privacy and retention rules, human answer ratings, a severity-aware incident process, and a predefined kill-switch owner.

4. Requirements: Functional Behavior

The PRD turns the discovery findings into observable functions. A ticket request carries an ID, channel, subject, body, and customer metadata. The API must validate and normalize supported inputs while preserving message meaning. Intent and urgency are separate predictions: each returns its selected label, numeric score, and alternatives, with no assumption that an alternative list is a retrieval result.

Requirement group	Acceptance evidence in this prototype
Normalize / classify	/ticket schema and response; intent and urgency alternatives; malformed input handled safely.
Retrieve approved evidence	Ranked source chunks with document and chunk IDs, text, and relevance scores.
Route / escalate	Deterministic action and reason; uncertainty and sensitive intents go to human review.
Draft / validate	Optional provider-backed draft or local fallback; citation, grounding, PII, integrity, and scope checks can block.
Persist / audit	SQLite decision with scores, sources, thresholds, reason, guardrail outcomes, and latency.
Batch evaluation	Input and output paths are command-line arguments; one item is processed per ticket.
Emergency stop	Filesystem kill-switch routes would-be automatic replies to humans.

The service supports one ticket at a time at the API boundary. The evaluation harness loops over a collection and invokes the service for each ticket. It does not mix multiple tickets into one retrieval or ask the classifier to jointly assess a batch. This separation makes each decision traceable to one ticket.

4. Requirements: Quality, Measures, and Traceability

Quality requirements include p95 local processing under three seconds, safe fallback when a provider is unavailable, no committed credentials, auditable decision records, and confidence calibration within five percentage points. Before a pilot, the evaluation framework calls for at least 50 responses independently rated by two assessors, with hallucination at most five percent and citation accuracy at least 95 percent. The rating study has not been performed.

The November 8 local p95 of 3.67 milliseconds is below the computational target for the service path measured by the harness. It excludes provider latency, network and queueing, human wait, and the time until a substantive customer reply. It is therefore not evidence that CloudServe meets its customer response promise.

Quality claim	Current evidence	Status
Intent / urgency prediction	80 labeled tickets in visible validation	Intent strong in-sample; urgency weak
Confidence calibration	ECE 16.64 pp, target <=5 pp	Target not met
Retrieval	49/53 expected-source hits	Proxy only; human relevance open
Answer quality / citations	33 generated responses; automatic checks only	Two-assessor audit outstanding
Privacy / secrets	Example config and source inspection	No production retention policy; history scanned
Reliability / availability	Offline run only	99.5% uptime cannot be established

Requirement status is evidence-specific. “Implemented” means a code path exists, not that the business threshold has passed. The 12 acceptance behaviors are summarized in Appendix C.

5. Architecture and Design: End-to-End Flow

A single API request follows an inspectable path. FastAPI validates the payload and normalizes channel and text fields. The classifier predicts intent and urgency separately. The retrieval layer searches the approved article corpus and returns the highest scoring chunks with stable document references. A deterministic policy evaluates labels, score floors, risk categories, evidence availability, and the global kill switch.

If the gates allow a response, the generator can call an optional Groq or OpenRouter chat-completions endpoint with the ticket and retrieved excerpts. If no provider is configured, local fallback behavior uses the available source content or routes to a human according to the policy. A deterministic validator checks the draft and citations before it is returned. Each path stores a decision record in SQLite.

The browser chat is a thin client for POST /ticket. It shows the answer when action is auto_respond and a human-review message otherwise. The API response keeps detailed predictions, alternatives, sources, escalation summary, guardrail results, latency, and decision ID for integrations and audit. The batch harness calls the same service once per input ticket and writes aggregate metrics and per-ticket decisions.

Layer	Implementation	Boundary
Ingress	FastAPI /ticket	One ticket per request
Prediction	scikit-learn intent and urgency classifiers	No LLM classification
Retrieval	TF-IDF vectors persisted in Chroma	Similarity is not a calibrated probability
Policy / validation	Deterministic Python rules	Fail closed to escalation/block
Drafting	Optional Groq/OpenRouter or local fallback	Provider path requires credentials; not live-verified
Audit	SQLite decision log	Local prototype, not production retention system

5. Architecture and Design: Retrieval and Classification

The documentation corpus is divided into fixed character windows of 800 characters with 120 characters of overlap. TF-IDF vectorization creates lexical vectors and Chroma persists and ranks them locally. The setup guide proposes sentence-transformer embeddings; this implementation selected TF-IDF to avoid model downloads and keep retrieval deterministic. The repository includes a development-only ablation comparing Markdown sections with the fixed-window strategy.

In the development ablation, fixed windows produced expected-document hit@3 of 93.3% versus 92.7% for section chunks, and top-1 of 86.8% versus 81.5% on 357 answerable labeled tickets. This proxy favored fixed windows for the tested corpus; it does not prove semantic relevance or answer support. The November 8 validation run hit an expected source on 49 of 53 eligible records (92.45%).

Intent and urgency are independent scikit-learn models trained on development labels. Their scores are returned separately. The top-ranked alternative is another class prediction; it does not represent RAG evidence. A retrieval score is computed from the vector search and should not be conflated with model confidence. Each threshold is a policy setting that determines whether to continue to answer or escalate.

The label sample is small and highly structured. Perfect observed intent accuracy is not generalization evidence. Urgency accuracy remains below half on the visible 80-ticket set. The observed intent ECE is 16.64 percentage points, so returned confidence should not be described as a calibrated guarantee. A larger, untouched calibration dataset and reliability analysis are prerequisites to broad automation.

5. Architecture and Design: Routing, Generation, and State

The router returns one of three actions: auto_respond, escalate, or blocked. Confidence floors are evaluated separately for intent and urgency. Risk categories such as security, compliance, data residency, unclear requests, and account-sensitive changes are handled conservatively. The decision reason explains the threshold or policy that led to the action.

Generation is optional and is not used for intent or urgency prediction. When configured, the provider receives only the ticket context and selected approved passages plus prompt instructions. Environment variables hold provider configuration. The implementation uses standard-library HTTP rather than the OpenAI SDK. This report's clean-checkout run used the local configuration and does not claim a successful request to an external provider.

The validator checks personally identifying information, instruction integrity, grounding/citations, and response tone/scope. It relies on deterministic and lexical heuristics. A passing citation reference proves that an ID resolves to a stored source; it does not prove each claim is semantically supported. An automated pass is therefore a filter result, not a human quality rating.

SQLite stores one record per processed ticket: normalized fields, predictions, score alternatives, sources, policy thresholds, action/reason, guardrail outcomes, version identifiers, latency, and errors. The local database makes the prototype auditable. Real customer text would require access control, encryption/backups, retention and deletion rules, and incident ownership before use.

6. Implementation: API, Browser, and Evaluation Harness

The API accepts one ticket and returns a structured decision. The browser chat sends the same request and keeps the customer-facing behavior simple: show the answer for auto_respond; otherwise state that the ticket has been sent for human review. The more detailed response remains available to the caller and decision log. Input is rendered as text in the interface, not interpreted as HTML.

The /tickets compatibility route aliases the one-ticket behavior. /health checks initialization, /metrics exposes Prometheus counters and histograms, and /docs exposes OpenAPI documentation. The service can start with local fallback and does not require a paid LLM to classify, retrieve, or evaluate.

The harness accepts --input and --output, processes a list of schema-compatible ticket records, writes per-ticket decisions and aggregate metrics, and records dataset/source hashes and invocation metadata. It also reconciles harness decisions with SQLite decision IDs. The hidden assessment input should be processed only by the evaluator; the local package includes the supplied visible set and has not used a hidden file.

The README documents installation, environment values, API usage, provider check, evaluation, threshold audit, retrieval ablation, monitoring and test commands. The delivered repository includes source modules, prompt files, test fixtures, CI configuration, architecture/governance notes, evaluation scripts, and the supplied capstone dataset and instructions.

6. Implementation: Tests and Reproducibility

The test suite was run from a fresh clone of the repository snapshot committed as d285977 on 8 November 2026. Following the README setup with Python 3.14.5 and the pinned requirements, all 16 tests passed. Coverage includes channel normalization, deterministic routing, guardrail behavior, generator fallback and retry, logged decisions, API behavior, kill switch, harness metrics, and ticket-ID-matched ground-truth checks.

Fresh-checkout verification	Observed result
Source revision	d285977 (snapshot parented to project commit 8f64256)
Environment	Python 3.14.5; README install steps; pinned requirements
Test command	python -m pytest tests/ -v
Tests	16 passed; one dependency deprecation warning
Unattended harness	80 tickets processed in one invocation
Decision persistence	80 of 80 decisions reconciled
Live provider / hidden evaluation	Not run; provider is optional, hidden set unavailable

The fresh clone resolved the prior verification gap: the submitted source folder is now an independent Git repository with its own history, and the same README test command succeeds from a new working directory. The final archive includes that .git directory. A local clean-checkout pass is not the hidden assessment gate; only the evaluator can run the withheld set.

7. Evaluation: Method and Measures

The primary run used the supplied validation_tickets.json and the packaged corpus from a fresh clone. The harness was invoked once on 8 November 2026 at 2026-11-08 16:09:41 UTC. It processed 80 tickets in one invocation and wrote decisions.json and metrics.json. The input SHA-256 was 94c5b1adc1203f06d59522a909d755e8671e9bf1b165a0bc5a74ab62501c827b; the source revision was d285977; the configured intent and urgency thresholds were 0.60 and 0.45.

Prediction metrics compare model outputs to supplied labels. Retrieval hit rate is the share of eligible answerable tickets for which at least one expected document appears in returned sources. Route accuracy compares action with expected-route labels. ECE summarizes the weighted gap between stated confidence and empirical correctness across bins. Local latency measures service processing, not provider or customer response time.

The validation file contains 80 tickets and is visible during development, so repeated runs are not the hidden assessment. The report retains historical results from November 2 and November 7 to show the routing/evaluation changes. The November 8 packaged clean-checkout execution is the primary current result. A separate working-copy smoke run earlier on November 8 used a different intent threshold (0.55 rather than the packaged 0.60); it produced the same action counts and classification/retrieval metrics and is not the primary report run.

The hidden 120-ticket assessment set was not present in the supplied workspace and was not run. No report claim is made about hidden-set performance.

7. Evaluation: November 8 Clean-Checkout Results

Metric	Observed result	Interpretation boundary
Actions	33 auto (41.25%); 47 escalate (58.75%); 0 block	Action share, not FCR
Intent	80/80 accuracy (100%)	Small structured visible sample
Urgency	35/80 accuracy (43.75%)	Weak performance; keep triage supervised
Expected source hit	49/53 (92.45%)	Document-level proxy, not semantic support
Route accuracy	51/80 (63.75%)	Agreement with expected route labels
Intent ECE	16.64 pp; target <=5 pp	Calibration target not met
Local latency	Mean 3.29; median 3.23; p95 3.55 ms	Excludes provider, network and queue time
SQLite log coverage	80/80 (100%)	Local persistence test succeeded

All 33 generated responses had citation identifiers resolving to stored passages, giving a 100% reference-resolution proxy. This does not establish claim-level citation accuracy. The run produced zero grounding blocks; that is an observed validator outcome for this run, not evidence that all responses are semantically safe.

Intent’s perfect score is constrained by the synthetic sample and class supports. Urgency performance is materially weaker: 45 of 80 urgency labels differ from the prediction. Confidence calibration also misses its target. These figures justify retaining human oversight and avoiding claims that confidence alone establishes answer correctness.

7. Evaluation: Business Evidence and Run History

The same validation records include historical FCR of 46.25%, repeat contact of 21.25%, and mean historical satisfaction of 2.6/5. These are recorded outcomes attached to the supplied ticket records, not ratings or resolutions caused by the system. The evaluator has no post-response outcome event, and there are no customer-visible first-reply timestamps in this run.

Visible run	Thresholds (intent / urgency)	Auto / escalate / block	Retrieval hit	Context
2 Nov 2026	Archived report does not state	8 / 65 / 7	51/53 (96.23%)	Earlier package metrics; p95 6.49 ms
7 Nov 2026	0.60 / 0.45	33 / 47 / 0	49/53 (92.45%)	One invocation; p95 7.86 ms
8 Nov 2026 working copy	0.55 / 0.45	33 / 47 / 0	49/53 (92.45%)	Preliminary smoke run; p95 3.62 ms
8 Nov 2026 clean clone (initial)	0.60 / 0.45	33 / 47 / 0	49/53 (92.45%)	Before .env copy; p95 3.67 ms
8 Nov 2026 clean checkout (final)	0.60 / 0.45	33 / 47 / 0	49/53 (92.45%)	Primary; p95 3.55 ms; d285977

Run counts are not to be collapsed into a claim that every result came from one invocation. The November 8 primary run was one harness invocation after completing the README setup; an earlier clean-clone check used the same defaults before copying .env. Earlier visible validation runs are kept as historical comparisons; they are not hidden-set runs. The November 2 metrics did not contain harness invocation metadata, and its count is described as one only because the earlier report explicitly records that run as once.

The current run reports 100% observed intent accuracy and 43.75% urgency accuracy, but the small sample, label construction, and repeated visible-set exposure limit inference. This evaluation is a reproducibility and prototype-behavior check, not an independent final benchmark.

8. Governance and Risk: Controls and Ownership

The prototype's policy is intended to fail closed. Low confidence, sensitive request classes, weak source evidence, detected personal data, or integrity/scope failures should result in escalation or a blocked draft. A filesystem kill switch converts would-be automatic replies to human review. Every processed ticket records the reason, source IDs, thresholds and guardrail outcomes for later investigation.

Risk	Severity	Current control	Remaining evidence
Unsupported answer	High	Source-based drafting; lexical grounding check; conservative route	Two-assessor review of at least 50 answers
PII in response	Critical	Deterministic PII check; local test cases	Production privacy, access, retention and deletion controls
Instruction injection	High	Treat ticket text as untrusted; integrity guardrail	Adversarial testing on larger set
Urgency error	High	Separate urgency model and human path	43.75% accuracy requires continued human triage
Provider outage	Medium	Timeout/rate-limit fallback path	Live provider request and outage verification
Data retention	High	Local SQLite decision log	Named owner and retention schedule absent

Before customer use, name a support owner, privacy contact, incident lead and escalation path. Define how to pause automation, preserve affected decision IDs, identify ticket and prompt/model versions, provide human correction, assess any exposure, and approve a monitored restart. The code path is not a substitute for an organizational incident procedure.

8. Governance and Risk: Fairness, Monitoring, and Human Review

The validation run reports automation coverage by customer tier, region, and language fluency, alongside historical FCR. The largest coverage range is 33.9 percentage points across regions; the historical FCR ranges are 24.4 points by tier, 23.8 by region, and 33.0 by fluency. Groups are small (enterprise n=8, Latin America n=7, non-fluent n=19). These are descriptive dataset differences, not observed system response-quality gaps or causal effects.

A fairness conclusion needs independently rated quality, adequate sample counts, uncertainty intervals, and a review of severity-weighted errors. The current 33 automatic drafts were not rated by two independent assessors. There are no live customer outcome measures, no temporal service window, and no response time series. Thus the 99.5% availability goal and response-time target cannot be asserted from one offline run.

Prometheus metrics cover action, channel, guardrail events, confidence and service latency. The supplied dashboard is a starter configuration. An operational monitor should track routing distribution, queue impact, error/fallback rates, response quality, subgroup outcomes, and the kill-switch state. Threshold alerts need an assigned on-call owner and documented action.

Before a pilot, two assessors should independently review at least 50 answers for factual support, citation correctness, completeness, action safety, tone and privacy, then report agreement and resolve disagreements. The sample should cover answer, escalation and blocked outcomes and include risk and language slices. This audit is not complete.

9. Requirements Revision: Evidence-Driven Changes

The revision log records changes prompted by implementation and visible evaluation. Urgency was separated from intent and is not treated as reliable autonomous priority because observed accuracy remains 43.75%. The broad requirement “answer when documentation exists” was narrowed to “answer only when classifier, source, policy, and validator gates pass”; the latest set still produced 41.25% automatic responses, so the lower escalation target remains unmet.

The automated quality proxy was demoted from a quality claim to a filter signal. Resolvable citation IDs and lexical grounding checks do not substitute for human review. The evaluation requirement now emphasizes arbitrary input and output paths, hashes, run counts, and a separate visible-versus-hidden distinction. Fairness reporting separates historical outcomes and action coverage from independently measured response quality.

Earlier assumption	Evidence from latest run	Revised position
A document match supports automation	49/53 expected-document hits, but no semantic rating	Use retrieval as evidence input, not answer approval
Classifier scores support safe autonomous routing	Intent ECE 16.64 pp; urgency accuracy 43.75%	Keep uncertain and high-risk tickets with people
Automated citation checks establish quality	33 references resolved; no human review	Report proxy only; require two assessors
Fast local latency means timely customer reply	p95 3.55 ms; no reply timestamps	Measure end-to-end lifecycle in a pilot
Visible dataset represents hidden assessment	80 visible tickets; hidden set unavailable	Preserve hidden-set isolation; evaluator runs once

The primary current report values now come from a clean-checkout run dated 8 November 2026. Earlier visible-set runs remain identified separately; their differing route thresholds and action outcomes are not silently blended.

10. Conclusions and Next Steps

The project delivers a locally runnable ticket decision service, a chat interface, retrieval over approved documents, an optional answer-generation path, deterministic checks, SQLite audit records, monitoring configuration, an evaluation harness, and a reproducible clean-checkout package. The 8 November clean-clone verification processed the full visible set, logged every decision, and passed all 16 automated tests.

The evidence does not support deployment. Urgency accuracy is weak, confidence calibration misses its target, generated response quality has not been independently rated, historical dataset outcomes are not system outcomes, and no live response-time or satisfaction data exist. The available evidence supports controlled development and human-supervised demonstration only.

Next steps, in order: (1) sign the effort log and record the remaining 17.5 scheduled hours as actual only after completing that work; (2) perform the required two-assessor audit of at least 50 outputs; (3) analyze urgency errors and recalibrate on a separate dataset; (4) exercise provider success, timeout and failure paths; (5) establish privacy, retention and incident ownership; (6) instrument a controlled pilot for first reply, resolution, repeats and CSAT; and (7) let the evaluator run the withheld assessment exactly once.

The submission deadline shown in this report is November 13, 2026, per the student’s correction. The package check is scheduled for that date and is not claimed to have occurred early. The required video remains pending and must demonstrate a success, an escalation, a guardrail block, and the unattended visible-set run.

Appendix A. Detailed November 8 Metrics

Primary artifact: validation_metrics_2026-11-08.json and validation_decisions_2026-11-08.json. The run used the packaged Git snapshot d285977 and the supplied 80-ticket visible validation input.

Area	Measure	Result
Volume	Processed / logged	80 / 80
Routing	Auto / escalate / blocked	33 / 47 / 0
Intent	Accuracy; macro precision / recall / F1	1.000; 1.000 / 1.000 / 1.000
Urgency	Accuracy	0.4375 (35 of 80)
Retrieval	Eligible tickets / expected hits / hit rate	53 / 49 / 0.9245
Routing	Accuracy against expected route	0.6375 (51 of 80)
Calibration	Intent ECE / target	0.1664 / <=0.05
Latency	Mean / median / p95 milliseconds	3.391 / 3.330 / 3.672
Historical outcomes	FCR / repeats / mean rating	46.25% / 21.25% / 2.60 of 5
First reply	Observed timestamped replies	0; mean and median unavailable
References	Citation IDs resolved	33 of 33; automated reference check only
Provider	External provider invocation	Not used / not verified live

Source hashes: input 94c5b1adc1203f06d59522a909d755e8671e9bf1b165a0bc5a74ab62501c827b; source 3821f6c9b5317867f7a754bd6ae4ee1f670edf90314183d33ea5ef64126c74c2. Harness start time: 2026-11-08 16:09:41 UTC. The generator-quality audit and hidden assessment are not included in these results.

Appendix B. Metric Definitions and Interpretation Limits

Metric	Calculation / source	Limit
Intent accuracy	Predicted intent equals label for labeled tickets	Small synthetic validation sample; not production generalization
Urgency accuracy	Predicted urgency equals supplied label	Does not show clinical/operational severity calibration
Expected document hit	At least one expected doc ID retrieved for eligible ticket	Does not prove chunk sufficiency or claim support
Route accuracy	System action compared with expected route	Not safe resolution, FCR or customer satisfaction
Auto-response coverage	Automatic responses divided by processed tickets	Not FCR or containment
ECE	Weighted confidence-versus-correctness gap across populated bins	Only two populated bands; noisy on 80 records
Latency	Service-side local timing in harness	Excludes provider/network/queue and agent response
Historical FCR / CSAT	Ticket dataset outcome fields	Not caused by this system; not ratings of generated drafts
Citation reference resolution	Generated citation IDs resolve to retrieved source IDs	Does not establish that cited text supports each claim
Fairness coverage	Action rates within metadata groups	Small groups; not response-quality parity

The clean-checkout output is retained in the report folder. Its decisions and aggregate metrics can be independently inspected. Any assessor should rerun only the hidden data provided for the assessment, not infer those results from this visible sample.

Appendix C. Clean-Checkout Verification and Traceability

Verification was performed on 8 November 2026 using a fresh local clone of the packaged repository. README setup commands were followed, dependencies installed from requirements.txt, and then the prescribed pytest and evaluation commands were run.

Acceptance area	Repository evidence	Verification state
Ticket API / normalization	src/api.py, src/schemas.py, src/service.py	Automated tests pass
Intent / urgency alternatives	src/classifier.py, tests/test_components.py	Tested; urgency quality 43.75% on visible data
Retrieval / citations	src/retrieval.py, src/chunking.py	49/53 expected doc hit; human relevance open
Routing / reasons	src/router.py, src/service.py	Visible run; route-label agreement 63.75%
Generation / safe fallback	src/generator.py, prompts/generation_v1.md	Fallback tests pass; live provider not verified
Guardrails / kill switch	src/validator.py, src/config.py	Automated tests pass; semantic audit open
Decision log	src/persistence.py	80/80 records reconciled in clean run
Batch harness	evaluation/harness.py	80 visible tickets in one invocation
Monitoring / CI	monitoring/, .github/workflows/tests.yml	Configuration included
History / fresh checkout	Git commit d285977; parent 8f64256	Clone clean; all 16 tests pass

Test command result: 16 passed, one dependency deprecation warning. Python 3.14.5 was used. The repository history includes the upstream project commits and the packaging snapshot. The history scan found no matches for common OpenAI, Groq, GitHub, or AWS credential formats; this pattern scan does not replace a complete secret scanner.